

On-Device Real-Time Speech Intelligence

gorch + Mimi — capabilities, evidence, and fit

Executive summary. We have built a speech-understanding pipeline that runs entirely on a single machine — no cloud services, no GPU, no Python — and processes live audio roughly nine times faster than real time on a MacBook CPU. It listens to speech, understands properties of it (what was said from a known vocabulary, who is speaking), and converts speech into the same compact token format that state-of-the-art voice AI systems use internally. Every capability claimed below is backed by an automated test that runs against real audio and real reference implementations.

What it is

gorch is our in-house machine-learning library written in Go: small, dependency-free, and tuned for Apple hardware. **Mimi** is an open-source speech codec from Kyutai (the lab behind the Moshi voice assistant); it compresses speech into a compact stream of numbers while preserving both the words and the voice. We ported Mimi's listening half natively into gorch and verified it against the original to near floating-point exactness. The result is one self-contained program that does what previously required a Python/PyTorch stack:

- **Ingestion** — live or recorded speech in, compact features out, in real time on CPU;
- **Understanding** — small classifiers trained on those features in seconds, on the same machine;
- **Tokenization** — speech converted to/verified against the discrete “speech token” format used by modern voice AI models.

The evidence

All numbers measured on a MacBook (Apple M4, CPU only) with automated tests that are part of the codebase and re-run on every change. “Reference” means the original PyTorch implementation.

Claim	Measured result	How it was proven
Faster than the reference implementation	283 ms to process 10 s of audio (reference: 334 ms)	Benchmark in the test suite
Real-time streaming with headroom	8.8 ms to process each 80 ms audio chunk (9× real-time); ~90 ms microphone-to-decision	Steady-state benchmark; streaming output matches batch output to $\sim 10^{-6}$
Faithful port of Mimi	Output matches the reference to $\sim 10^{-6}$ at every internal stage	Golden-fixture tests against reference activations at 5 pipeline stages
Understands spoken content	100% spoken-digit recognition; 97% on a speaker never seen in training	3,000-recording public dataset (FSDD), held-out test splits
Recognizes who is speaking	100% speaker identification (6 speakers)	Same dataset, held-out recordings
Works on real-world voices it never trained on	100% (30/30) on digits spoken by three modern synthetic voices	Azure OpenAI voice clips, independently verified by a speech recognizer before use
Produces valid speech tokens	Token-for-token identical to the reference; tokens decode back to speech a recognizer transcribes correctly 30/30	Exact-match test across all 32 codebooks + round-trip through the reference decoder and an independent ASR
Learns new tasks in seconds	Classifier training: ~2 s ; per-decision inference: ~2 μs	Timed in the e2e tests

Audible proof exists as well: a side-by-side recording of original speech versus speech reconstructed purely from gorch-produced tokens.

What this enables

Real-time call analytics (contact center / PBX)

Live language detection for routing, sentiment/emotion tracking during calls, and keyword or compliance-phrase spotting — with decisions every 80 ms at ~90 ms latency, on the call server itself. The paths are proven; production heads need task-specific training data (days, not months).

Privacy-critical, on-premise voice AI

Nothing leaves the machine: no cloud APIs, no Python runtime, one Go binary plus one 380 MB model file. Attractive wherever GDPR, healthcare, or client-confidentiality rules make cloud speech processing hard to approve — and there is no per-minute API cost.

Voice interfaces on modest hardware

Command-word and small-vocabulary recognition (the digit tests are exactly this) with 9× real-time headroom on a laptop CPU — several audio streams per machine, GPU left free for other work.

Speaker identification in known groups

Who in a meeting or on a known team line is speaking — 100% on the 6-speaker benchmark; features demonstrably carry voice identity.

Foundation for speech-LM products (research)

gorch now speaks the token language of Moshi-class voice AI. Combined with gorch's existing GPT training stack, this opens in-house experimentation with speech language models — understanding and, longer-term, generation.

What it is not (today)

Not a general transcription engine. It recognizes known vocabularies and speech properties; open “type what was said” transcription is a different, much larger model class (Whisper-scale). Those models can run alongside it; porting one into gorch would be a separate project.

Not a voice generator. We produce and verify speech tokens, but turning tokens back into audible speech currently uses the reference decoder (Python). A native decoder port is a known, bounded follow-up if voice output becomes a goal.

Not for music or high-fidelity audio. Mimi is a speech codec operating at ~1.1 kbit/s — reconstructions are clearly intelligible but audibly compressed. It analyzes speech; it is not a hi-fi audio tool.

Apple-hardware only, for now. The code targets macOS/Apple Silicon (Apple's Accelerate framework). A Linux/server port is feasible — the design isolates the hardware layer — but is unscheduled work.

Sentiment/language models not yet trained. The pipeline and its capacity for these tasks are proven (speaker identity — the same signal family — survives at 100%), but production language-ID and emotion heads still need training on labeled data.

How it was built and verified

The port was executed as a staged, test-gated plan: every stage of the Go implementation is compared against reference outputs recorded from the original implementation (“golden fixtures”), streaming is verified equivalent to batch processing, and two end-to-end tests re-prove the headline accuracies on every run — including one on speech generated by third-party (Azure) voices and independently verified by a speech recognizer. The work is on the main branch with the full test suite green.

Sources: [e2e/mimi_fsdd_test.go](#), [e2e/mimi_native_fsdd_test.go](#), [e2e/mimi_realworld_test.go](#), [audio/mimi/*](#) golden tests and benchmarks, [doc/mimi-audio-report.md](#), [doc/plans/0006-mimi-native-encoder.md](#). Hardware: Apple M4, 24 GB, CPU only. Mimi: [kyutai/mimi](#) (CC-BY 4.0 weights).